

General Mathematics

Concepts, Methods and Practice

Topics Covered:

- Number Systems
- Fractions, Decimals and Percentages
- Algebra
- Geometry
- Statistics and Probability
- Ratios and Proportions
- Coordinate Geometry
- Trigonometry
- Problem Solving Strategies

Chapter 1: Number Systems

Numbers are the foundation of all mathematics. Different sets of numbers are used depending on the context.

Types of Numbers

- **Natural Numbers** – Positive counting numbers: 1, 2, 3 ...
- **Whole Numbers** – Natural numbers plus zero: 0, 1, 2, 3 ...
- **Integers** – Whole numbers and their negatives: ... -2, -1, 0, 1, 2 ...
- **Rational Numbers** – Expressible as p/q where q is not zero.
- **Irrational Numbers** – Cannot be expressed as a fraction (e.g., π , $\sqrt{2}$).
- **Real Numbers** – All rational and irrational numbers combined.

Number Properties

- **Commutative:** $a + b = b + a$ and $a \times b = b \times a$
- **Associative:** $(a + b) + c = a + (b + c)$
- **Distributive:** $a \times (b + c) = ab + ac$
- **Identity:** Adding 0 or multiplying by 1 leaves a number unchanged.

Prime and Composite Numbers

A **prime number** has exactly two factors: 1 and itself (2, 3, 5, 7, 11 ...). A **composite number** has more than two factors. The number 1 is neither prime nor composite.

Chapter 2: Fractions, Decimals and Percentages

These three representations are interchangeable and appear constantly in everyday mathematics.

Fractions

A fraction a/b represents a parts of a whole divided into b equal parts.

- **Addition/Subtraction:** Find a common denominator, then add/subtract numerators.
- **Multiplication:** Multiply numerators together and denominators together.
- **Division:** Multiply by the reciprocal of the divisor.

Decimals

Decimals are fractions with denominators of powers of 10. To convert a fraction to a decimal, divide the numerator by the denominator. For example, $3/4 = 0.75$.

Percentages

Percentage means 'per hundred'. To convert a fraction to a percentage, multiply by 100.

Percentage of a quantity: $(\text{Percent} / 100) \times \text{Whole}$

Percentage change: $[(\text{New} - \text{Old}) / \text{Old}] \times 100$

Finding the original: $\text{Amount} / (\text{Percent} / 100)$

Chapter 3: Algebra

Algebra uses letters (variables) to represent unknown quantities, allowing us to write general rules and solve equations.

Key Concepts

- **Expression** – A combination of numbers, variables, and operations (e.g., $3x + 5$).
- **Equation** – Two expressions set equal to each other (e.g., $2x + 3 = 11$).
- **Variable** – A letter representing an unknown value.
- **Coefficient** – The numerical factor of a term (in $4y$, the coefficient is 4).

Solving Linear Equations

Isolate the variable by applying inverse operations to both sides equally.

Solve: $3x - 7 = 14$

Step 1: Add 7 to both sides $\rightarrow 3x = 21$

Step 2: Divide both sides by 3 $\rightarrow x = 7$

Expanding and Factorising

- **Expanding**: Apply the distributive law. E.g., $2(x + 3) = 2x + 6$.
- **Factorising**: Reverse of expanding. E.g., $6x + 9 = 3(2x + 3)$.

Quadratic Equations

Form: $ax^2 + bx + c = 0$. Solve by factorising, completing the square, or the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Chapter 4: Geometry

Geometry deals with shapes, sizes, and spatial relationships. It has applications in architecture, engineering, art, and navigation.

Angles

- **Acute:** Less than 90 degrees
- **Right:** Exactly 90 degrees
- **Obtuse:** Between 90 and 180 degrees
- **Straight:** 180 degrees
- **Reflex:** Between 180 and 360 degrees

Key Formulas — 2D Shapes

Shape	Area	Perimeter
Rectangle	length x width	$2(l + w)$
Triangle	$(\text{base} \times \text{height}) / 2$	$a + b + c$
Circle	$\pi \times r^2$	$2 \times \pi \times r$
Parallelogram	base x height	$2(a + b)$
Trapezoid	$(a + b) / 2 \times h$	$a + b + c + d$

3D Shapes

- **Cube:** Volume = s^3 , Surface Area = $6s^2$
- **Cuboid:** Volume = $l \times w \times h$, Surface Area = $2(lw + lh + wh)$
- **Sphere:** Volume = $(4/3) \times \pi \times r^3$, Surface Area = $4 \times \pi \times r^2$
- **Cylinder:** Volume = $\pi \times r^2 \times h$, Curved Surface Area = $2 \times \pi \times r \times h$

Chapter 5: Statistics and Probability

Statistics involves collecting, organising, and interpreting data. Probability measures the likelihood of events.

Measures of Central Tendency

- **Mean** – Sum of all values divided by the count of values.
- **Median** – The middle value when data is sorted in order.
- **Mode** – The most frequently occurring value.

Measures of Spread

- **Range** – Difference between the maximum and minimum values.
- **Variance** – Average of squared differences from the mean.
- **Standard Deviation** – Square root of variance; shows spread around the mean.

Probability

Probability = (Favourable outcomes) / (Total possible outcomes). Ranges from 0 (impossible) to 1 (certain). For complementary events: $P(A) + P(\text{not } A) = 1$.

Example: Probability of rolling a 4 on a standard die = $1/6$ = approximately 0.167 (16.7%).

Chapter 6: Ratios and Proportions

A ratio compares two quantities. A proportion states that two ratios are equal. These concepts underpin cooking, scale drawings, currency exchange, and more.

Ratios

A ratio $a:b$ means for every a units there are b units. Ratios simplify like fractions. E.g., $12:8$ simplifies to $3:2$.

Dividing in a Given Ratio

Divide 60 in the ratio 2:3:
Total parts = $2 + 3 = 5$
Each part = $60 / 5 = 12$
Shares: $2 \times 12 = 24$ and $3 \times 12 = 36$

Direct and Inverse Proportion

- **Direct proportion** – As one quantity increases, the other increases at the same rate. $y = kx$.
- **Inverse proportion** – As one quantity increases, the other decreases. $y = k/x$.

Scale and Maps

A map scale of 1:50,000 means 1 cm on the map equals 500 m in reality. Ratios are essential for scaling blueprints, models, and geographic representations.

Chapter 7: Coordinate Geometry

Coordinate geometry combines algebra and geometry using a coordinate plane to represent and analyse geometric shapes.

The Coordinate Plane

The Cartesian plane has a horizontal x-axis and vertical y-axis intersecting at the origin (0,0). Any point is written as (x, y). The plane is divided into four quadrants.

Key Formulas

Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Midpoint: $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$

Gradient (slope): $m = (y_2 - y_1) / (x_2 - x_1)$

Equation of a line: $y = mx + c$ (m = slope, c = y-intercept)

Parallel and Perpendicular Lines

- **Parallel lines** have equal gradients: $m_1 = m_2$.
- **Perpendicular lines** have gradients that multiply to -1: $m_1 \times m_2 = -1$.

Chapter 8: Trigonometry

Trigonometry studies relationships between angles and sides of triangles. Used widely in physics, engineering, and navigation.

Trigonometric Ratios (Right-Angled Triangle)

sin(theta) = Opposite / Hypotenuse

cos(theta) = Adjacent / Hypotenuse

tan(theta) = Opposite / Adjacent

Memory aid: SOH – CAH – TOA

Common Angle Values

Angle	sin	cos	tan
0 deg	0	1	0
30 deg	1/2	sqrt3/2	1/sqrt3
45 deg	sqrt2/2	sqrt2/2	1
60 deg	sqrt3/2	1/2	sqrt3
90 deg	1	0	undefined

The Pythagorean Theorem

In any right-angled triangle: $a^2 + b^2 = c^2$, where c is the hypotenuse. This theorem connects algebra and geometry and is the basis of the distance formula.

Chapter 9: Problem Solving Strategies

Strong mathematical problem solving requires both knowledge of techniques and disciplined thinking. These strategies apply across all areas of mathematics.

The IDEAL Framework

- **I**dentify the problem clearly — what is actually being asked?
- **D**efine the information — what data is given and what is unknown?
- **E**xplore strategies — which method or formula applies here?
- **A**ct on the plan — carry out the solution step by step.
- **L**ook back — does the answer make sense in context?

Common Strategies

- Draw a diagram to visualise the problem.
- Work backwards from the desired result.
- Simplify by trying a smaller version of the problem.
- Look for patterns in numbers or shapes.
- Estimate first to check whether your answer is reasonable.

Summary: Mathematics is a universal language built on logic and pattern. Mastering its core concepts enables you to model, analyse, and solve problems across every field of human endeavour.